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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science (Co-op)

Sept 2024 – Present

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management, Linear Algebra, Functional Programming.

WORK EXPERIENCE

PengYang Asset Management Co., Ltd.

Beijing, China

Software Engineering Intern

May 2025 – Sept 2025

- Engineered a Python-based simulation engine to process **large-scale financial datasets**, optimizing portfolio distribution logic for performance.
- Automated daily risk reporting pipelines using SQL and Python scripts, reducing manual data entry time by approx. **10 hours/week**.
- Developed and backtested a predictive algorithm on historical datasets, improving model accuracy by 2% through rigorous parameter tuning.
- Implemented data validation checks and error handling for internal bond inventory tools to ensure data integrity for downstream users.
- Collaborated with analysts to translate complex mathematical requirements into production-ready Python code.

PROJECTS

Distributed Chess Engine (AlphaZero Clone) | Python, PyTorch, Multiprocessing

Jan 2025 – Present

- Architected a distributed AlphaZero-style engine using an **18-layer ResNet in PyTorch**, achieving a **1000 ELO increase** over **2000** iterations of self-play training.
- Accelerated Monte Carlo Tree Search (MCTS) by **400%** via **multiprocessing**, linearly scaling simulation throughput across **128 CPU cores**.
- Engineered a high-throughput data pipeline generating over **5,000** games per hour, utilizing **binary serialization** to minimize disk I/O overhead.
- Integrated a OneCycleLR scheduler to optimize neural network convergence speed during training cycles.

Full-Stack Analytics Dashboard | React, FastAPI, Docker, CI/CD

Jan 2025 – Present

- Developed a scalable microservices application to visualize GitHub repository trends with sub-second API latency.
- Built a high-performance REST API using **FastAPI** to ingest and transform live data, serving a **React** frontend.
- Containerized the entire stack using **Docker** to ensure consistent environments across delivery.
- Implemented a CI/CD pipeline using GitHub Actions to automate testing and deployment workflows.

Large Language Model (LLM) Inference Engine | C++, CUDA, CMake

Jan 2026 – Present

- Architected a bare-metal inference runtime for **Llama-3 architectures**, utilizing **C++20** and **CUDA** for hardware acceleration on an RTX 6000.
- Engineered custom CUDA kernels for **RoPE**, **RMSNorm**, and **SwiGLU** activations, bypassing high-overhead PyTorch abstractions to reduce latency.
- Implemented **4-bit Integer Quantization (INT4)** with a custom de-quantization kernel, enabling a **120B parameter model** to fit entirely within **98GB of VRAM**.
- Optimized memory hierarchy using a linear arena allocator and **KV-Caching** with paged attention, minimizing redundant computations during autoregressive generation.

TECHNICAL SKILLS

Languages: C++ (C++20), CUDA, Python, Java, SQL, TypeScript, Bash, HTML/CSS

Machine Learning: PyTorch, NumPy, Pandas, Transformers, HuggingFace, Scikit-learn

High Performance Computing: Multi-threading, OpenMP, SIMD, Quantization (INT4/8), Memory Management

Web and Backend: FastAPI, React, Node.js, Docker, PostgreSQL, REST APIs, Microservices

Developer Tools: Git, Linux/Unix, CMake, GDB, Valgrind, GitHub Actions, CI/CD

Concepts: Large Language Models (LLMs), Deep Learning, Reinforcement Learning, Distributed Systems, OOP